

PILOT-OFFICE-01 - 4-Story Office Frame

PILOT-OFFICE-01 Office Building Elastic Analysis Report

Purpose	Program smoke verification - draft, not for construction
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Engineer	S-Structures
Reviewer	Independent/manual verification pending
Package Version	m42-calculation-package

This report traces the current elastic analysis and preliminary member checks. It is not a sealed final design package.

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Design Basis And Loads

Nodes 45	Members 84	Load cases 6	Combos 28
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Case	Type	Loads	Fx	Fy	Fz
D	dead	48	0.000 kN	0.000 kN	-2016 kN
L	live	48	0.000 kN	0.000 kN	-840.00 kN
WX	wind	36	64.80 kN	0.000 kN	0.000 kN
WY	wind	36	0.000 kN	77.76 kN	0.000 kN
EX	seismic	36	175.85 kN	0.000 kN	0.000 kN
EY	seismic	36	0.000 kN	175.85 kN	0.000 kN

Story	Area	D total	L total	Wind X	Wind Y
1	120.00 m2	504.00 kN	240.00 kN	16.20 kN	19.44 kN
2	120.00 m2	504.00 kN	240.00 kN	16.20 kN	19.44 kN
3	120.00 m2	504.00 kN	240.00 kN	16.20 kN	19.44 kN
4	120.00 m2	504.00 kN	120.00 kN	16.20 kN	19.44 kN

Load Derivation Formula Trace

ID	Case	Formula	Inputs	Result
basis-occupancy	-	occupancy preset lookup	occupancy=office	Office
basis-plan-area	-	max(model width * model depth, 1)	Bx=12.00 m, By=10.00 m	120.00 m2
basis-roof-area	-	roofArea input or typical floor area	roofArea=120.00 m2	120.00 m2
D-ST1	D	A * qD	A=120.00 m2, qD=4.200 kN/m2	504.00 kN
L-ST1	L	A * qL	A=120.00 m2, qL=2.000 kN/m2	240.00 kN
WX-ST1	WX	pWX * By * h	pWX=0.450 kN/m2, By=10.00 m, h=3.600 m	16.20 kN
WY-ST1	WY	pWY * Bx * h	pWY=0.450 kN/m2, Bx=12.00 m, h=3.600 m	19.44 kN
D-ST2	D	A * qD	A=120.00 m2, qD=4.200 kN/m2	504.00 kN
L-ST2	L	A * qL	A=120.00 m2, qL=2.000 kN/m2	240.00 kN
WX-ST2	WX	pWX * By * h	pWX=0.450 kN/m2, By=10.00 m, h=3.600 m	16.20 kN
WY-ST2	WY	pWY * Bx * h	pWY=0.450 kN/m2, Bx=12.00 m, h=3.600 m	19.44 kN

ID	Case	Formula		Inputs	Result
D-ST3	D	$A * qD$		A=120.00 m2, qD=4.200 kN/m2	504.00 kN
L-ST3	L	$A * qL$		A=120.00 m2, qL=2.000 kN/m2	240.00 kN
WX-ST3	WX	$pWX * By * h$	pWX=0.450 kN/m2, By=10.00 m, h=3.600 m		16.20 kN
WY-ST3	WY	$pWY * Bx * h$	pWY=0.450 kN/m2, Bx=12.00 m, h=3.600 m		19.44 kN
D-ST4	D	$A * qD$		A=120.00 m2, qD=4.200 kN/m2	504.00 kN
L-ST4	L	$A * qL$		A=120.00 m2, qL=1.000 kN/m2	120.00 kN
WX-ST4	WX	$pWX * By * h$	pWX=0.450 kN/m2, By=10.00 m, h=3.600 m		16.20 kN
WY-ST4	WY	$pWY * Bx * h$	pWY=0.450 kN/m2, Bx=12.00 m, h=3.600 m		19.44 kN
EX-BASE	EX	$CsX * \text{sum}(Wi)$		CsX=0.079 g, sumW=2226 kN	175.85 kN
EY-BASE	EY	$CsY * \text{sum}(Wi)$		CsY=0.079 g, sumW=2226 kN	175.85 kN
EX-ST1	EX	$Vx * Wi * zi / \text{sum}(Wi * zi)$	Vx=175.85 kN, Wi=564.00 kN, zi=3.600 m		17.97 kN
EY-ST1	EY	$Vy * Wi * zi / \text{sum}(Wi * zi)$	Vy=175.85 kN, Wi=564.00 kN, zi=3.600 m		17.97 kN
EX-ST2	EX	$Vx * Wi * zi / \text{sum}(Wi * zi)$	Vx=175.85 kN, Wi=564.00 kN, zi=7.200 m		35.94 kN
EY-ST2	EY	$Vy * Wi * zi / \text{sum}(Wi * zi)$	Vy=175.85 kN, Wi=564.00 kN, zi=7.200 m		35.94 kN
EX-ST3	EX	$Vx * Wi * zi / \text{sum}(Wi * zi)$	Vx=175.85 kN, Wi=564.00 kN, zi=10.80 m		53.90 kN
EY-ST3	EY	$Vy * Wi * zi / \text{sum}(Wi * zi)$	Vy=175.85 kN, Wi=564.00 kN, zi=10.80 m		53.90 kN
EX-ST4	EX	$Vx * Wi * zi / \text{sum}(Wi * zi)$	Vx=175.85 kN, Wi=534.00 kN, zi=14.40 m		68.05 kN
EY-ST4	EY	$Vy * Wi * zi / \text{sum}(Wi * zi)$	Vy=175.85 kN, Wi=534.00 kN, zi=14.40 m		68.05 kN
D-DIST-ST1	D	$Dstory / \text{sum}(Lbeam)$	Dstory=504.00 kN, sumL=66.00 m, nBeam=12.00		7.636 kN/m
L-DIST-ST1	L	$Lstory / \text{sum}(Lbeam)$	Lstory=240.00 kN, sumL=66.00 m, nBeam=12.00		3.636 kN/m
WX-NODE-ST1	WX	$WXstory / nNodes$	WXstory=16.20 kN, nNodes=9.000		1.800 kN/node
WY-NODE-ST1	WY	$WYstory / nNodes$	WYstory=19.44 kN, nNodes=9.000		2.160 kN/node
EX-NODE-ST1	EX	$EXstory / nNodes$	EXstory=17.97 kN, nNodes=9.000		1.996 kN/node
EY-NODE-ST1	EY	$EYstory / nNodes$	EYstory=17.97 kN, nNodes=9.000		1.996 kN/node
D-DIST-ST2	D	$Dstory / \text{sum}(Lbeam)$	Dstory=504.00 kN, sumL=66.00 m, nBeam=12.00		7.636 kN/m
L-DIST-ST2	L	$Lstory / \text{sum}(Lbeam)$	Lstory=240.00 kN, sumL=66.00 m, nBeam=12.00		3.636 kN/m

ID	Case	Formula	Inputs	Result
WX-NODE-ST2	WX	WXstory / nNodes	WXstory=16.20 kN, nNodes=9.000	1.800 kN/node
WY-NODE-ST2	WY	WYstory / nNodes	WYstory=19.44 kN, nNodes=9.000	2.160 kN/node
EX-NODE-ST2	EX	EXstory / nNodes	EXstory=35.94 kN, nNodes=9.000	3.993 kN/node
EY-NODE-ST2	EY	EYstory / nNodes	EYstory=35.94 kN, nNodes=9.000	3.993 kN/node
D-DIST-ST3	D	Dstory / sum(Lbeam)	Dstory=504.00 kN, sumL=66.00 m, nBeam=12.00	7.636 kN/m
L-DIST-ST3	L	Lstory / sum(Lbeam)	Lstory=240.00 kN, sumL=66.00 m, nBeam=12.00	3.636 kN/m
WX-NODE-ST3	WX	WXstory / nNodes	WXstory=16.20 kN, nNodes=9.000	1.800 kN/node
WY-NODE-ST3	WY	WYstory / nNodes	WYstory=19.44 kN, nNodes=9.000	2.160 kN/node
EX-NODE-ST3	EX	EXstory / nNodes	EXstory=53.90 kN, nNodes=9.000	5.989 kN/node
EY-NODE-ST3	EY	EYstory / nNodes	EYstory=53.90 kN, nNodes=9.000	5.989 kN/node
D-DIST-ST4	D	Dstory / sum(Lbeam)	Dstory=504.00 kN, sumL=66.00 m, nBeam=12.00	7.636 kN/m
L-DIST-ST4	L	Lstory / sum(Lbeam)	Lstory=120.00 kN, sumL=66.00 m, nBeam=12.00	1.818 kN/m
WX-NODE-ST4	WX	WXstory / nNodes	WXstory=16.20 kN, nNodes=9.000	1.800 kN/node
WY-NODE-ST4	WY	WYstory / nNodes	WYstory=19.44 kN, nNodes=9.000	2.160 kN/node
EX-NODE-ST4	EX	EXstory / nNodes	EXstory=68.05 kN, nNodes=9.000	7.561 kN/node
EY-NODE-ST4	EY	EYstory / nNodes	EYstory=68.05 kN, nNodes=9.000	7.561 kN/node

KDS-Style Load Standard Audit

Symbol	Status	Mapped cases	Project input
D	mapped	D	Optional
L	mapped	L	Optional
Lr	missing	-	Optional
S	missing	-	Required
R	missing	-	Required
W	mapped	WX, WY	Required

Symbol	Status	Mapped cases	Project input
E	mapped	EX, EY	Required
H	missing	-	Required
F	missing	-	Required
T	missing	-	Required
EQUIPMENT	missing	-	Optional
CONSTRUCTION	missing	-	Optional
OTHER	missing	-	Optional

Elastic Analysis Summary

Status OK	Audit OK	Eq. residual 0.000	Max disp. 5.749 mm
Max util. 0.394	Warnings 0		

Combo	OK	Max disp.	Max util.	Residual
KDS-ST-01	OK	1.168 mm	0.124	0.000
KDS-ST-02	OK	1.436 mm	0.165	0.000
KDS-ST-04-WX-P	OK	2.118 mm	0.182	0.000
KDS-ST-04-WX-N	OK	2.118 mm	0.182	0.000
KDS-ST-04-WY-P	OK	2.253 mm	0.189	0.000
KDS-ST-04-WY-N	OK	2.253 mm	0.189	0.000
KDS-ST-05-EX-P	OK	5.749 mm	0.252	0.000
KDS-ST-05-EX-N	OK	5.749 mm	0.252	0.000
KDS-ST-05-EY-P	OK	5.289 mm	0.249	0.000
KDS-ST-05-EY-N	OK	5.289 mm	0.249	0.000
KDS-ST-06-WX-P	OK	1.851 mm	0.119	0.000
KDS-ST-06-WX-N	OK	1.851 mm	0.119	0.000
KDS-ST-06-WY-P	OK	2.005 mm	0.125	0.000
KDS-ST-06-WY-N	OK	2.005 mm	0.125	0.000
KDS-ST-07-EX-P	OK	5.654 mm	0.189	0.000
KDS-ST-07-EX-N	OK	5.654 mm	0.189	0.000
KDS-ST-07-EY-P	OK	5.189 mm	0.186	0.000
KDS-ST-07-EY-N	OK	5.189 mm	0.186	0.000
KDS-SVC-01	OK	0.834 mm	0.089	0.000
KDS-SVC-02	OK	1.106 mm	0.125	0.000
KDS-SVC-03-WX-P	OK	1.951 mm	0.146	0.000
KDS-SVC-03-WX-N	OK	1.951 mm	0.146	0.000
KDS-SVC-03-WY-P	OK	2.097 mm	0.153	0.000
KDS-SVC-03-WY-N	OK	2.097 mm	0.153	0.000
KDS-SVC-04-EX-P	OK	5.689 mm	0.216	0.000
KDS-SVC-04-EX-N	OK	5.689 mm	0.216	0.000
KDS-SVC-04-EY-P	OK	5.225 mm	0.213	0.000

Combo	OK	Max disp.	Max util.	Residual
KDS-SVC-04-EY-N	OK	5.225 mm	0.213	0.000

Advanced Elastic Trace

Item	Value
P-Delta	disabled
Modes	0
First period	0.000
RSA directions	0

Phase 5 Analysis Cases

No Phase 5 analysis cases defined.

Immutable Analysis Run Records

No Phase 7 analysis run records are attached.

Serviceability Drift Review

Combo	Story	Height	Drift	Ratio	D/L	Status
KDS-ST-01	1	3.600 m	0.005 mm	0.00000	0.000	OK
KDS-ST-01	2	3.600 m	0.006 mm	0.00000	0.000	OK
KDS-ST-01	3	3.600 m	0.006 mm	0.00000	0.000	OK
KDS-ST-01	4	3.600 m	0.021 mm	0.00001	0.001	OK
KDS-ST-02	1	3.600 m	0.007 mm	0.00000	0.000	OK
KDS-ST-02	2	3.600 m	0.008 mm	0.00000	0.000	OK
KDS-ST-02	3	3.600 m	0.005 mm	0.00000	0.000	OK
KDS-ST-02	4	3.600 m	0.024 mm	0.00001	0.001	OK
KDS-ST-04-WX-P	1	3.600 m	0.458 mm	0.00013	0.025	OK
KDS-ST-04-WX-P	2	3.600 m	0.583 mm	0.00016	0.032	OK
KDS-ST-04-WX-P	3	3.600 m	0.426 mm	0.00012	0.024	OK
KDS-ST-04-WX-P	4	3.600 m	0.255 mm	0.00007	0.014	OK
KDS-ST-04-WX-N	1	3.600 m	0.458 mm	0.00013	0.025	OK
KDS-ST-04-WX-N	2	3.600 m	0.583 mm	0.00016	0.032	OK
KDS-ST-04-WX-N	3	3.600 m	0.426 mm	0.00012	0.024	OK
KDS-ST-04-WX-N	4	3.600 m	0.255 mm	0.00007	0.014	OK
KDS-ST-04-WY-P	1	3.600 m	0.516 mm	0.00014	0.029	OK
KDS-ST-04-WY-P	2	3.600 m	0.636 mm	0.00018	0.035	OK

Combo	Story	Height	Drift	Ratio	D/L	Status
KDS-ST-04-WY-P	3	3.600 m	0.461 mm	0.00013	0.026	OK
KDS-ST-04-WY-P	4	3.600 m	0.268 mm	0.00007	0.015	OK
KDS-ST-04-WY-N	1	3.600 m	0.516 mm	0.00014	0.029	OK
KDS-ST-04-WY-N	2	3.600 m	0.636 mm	0.00018	0.035	OK
KDS-ST-04-WY-N	3	3.600 m	0.461 mm	0.00013	0.026	OK
KDS-ST-04-WY-N	4	3.600 m	0.268 mm	0.00007	0.015	OK
KDS-ST-05-EX-P	1	3.600 m	1.296 mm	0.00036	0.072	OK
KDS-ST-05-EX-P	2	3.600 m	1.859 mm	0.00052	0.103	OK
KDS-ST-05-EX-P	3	3.600 m	1.530 mm	0.00043	0.085	OK
KDS-ST-05-EX-P	4	3.600 m	0.950 mm	0.00026	0.053	OK
KDS-ST-05-EX-N	1	3.600 m	1.296 mm	0.00036	0.072	OK
KDS-ST-05-EX-N	2	3.600 m	1.859 mm	0.00052	0.103	OK
KDS-ST-05-EX-N	3	3.600 m	1.530 mm	0.00043	0.085	OK
KDS-ST-05-EX-N	4	3.600 m	0.950 mm	0.00026	0.053	OK
KDS-ST-05-EY-P	1	3.600 m	1.216 mm	0.00034	0.068	OK
KDS-ST-05-EY-P	2	3.600 m	1.696 mm	0.00047	0.094	OK
KDS-ST-05-EY-P	3	3.600 m	1.389 mm	0.00039	0.077	OK
KDS-ST-05-EY-P	4	3.600 m	0.855 mm	0.00024	0.048	OK
KDS-ST-05-EY-N	1	3.600 m	1.216 mm	0.00034	0.068	OK
KDS-ST-05-EY-N	2	3.600 m	1.696 mm	0.00047	0.094	OK
KDS-ST-05-EY-N	3	3.600 m	1.389 mm	0.00039	0.077	OK
KDS-ST-05-EY-N	4	3.600 m	0.855 mm	0.00024	0.048	OK
KDS-ST-06-WX-P	1	3.600 m	0.455 mm	0.00013	0.025	OK
KDS-ST-06-WX-P	2	3.600 m	0.580 mm	0.00016	0.032	OK
KDS-ST-06-WX-P	3	3.600 m	0.425 mm	0.00012	0.024	OK
KDS-ST-06-WX-P	4	3.600 m	0.248 mm	0.00007	0.014	OK
KDS-ST-06-WX-N	1	3.600 m	0.455 mm	0.00013	0.025	OK
KDS-ST-06-WX-N	2	3.600 m	0.580 mm	0.00016	0.032	OK
KDS-ST-06-WX-N	3	3.600 m	0.425 mm	0.00012	0.024	OK
KDS-ST-06-WX-N	4	3.600 m	0.248 mm	0.00007	0.014	OK
KDS-ST-06-WY-P	1	3.600 m	0.514 mm	0.00014	0.029	OK
KDS-ST-06-WY-P	2	3.600 m	0.635 mm	0.00018	0.035	OK
KDS-ST-06-WY-P	3	3.600 m	0.461 mm	0.00013	0.026	OK
KDS-ST-06-WY-P	4	3.600 m	0.263 mm	0.00007	0.015	OK

Combo	Story	Height	Drift	Ratio	D/L	Status
KDS-ST-06-WY-N	1	3.600 m	0.514 mm	0.00014	0.029	OK
KDS-ST-06-WY-N	2	3.600 m	0.635 mm	0.00018	0.035	OK
KDS-ST-06-WY-N	3	3.600 m	0.461 mm	0.00013	0.026	OK
KDS-ST-06-WY-N	4	3.600 m	0.263 mm	0.00007	0.015	OK
KDS-ST-07-EX-P	1	3.600 m	1.293 mm	0.00036	0.072	OK
KDS-ST-07-EX-P	2	3.600 m	1.856 mm	0.00052	0.103	OK
KDS-ST-07-EX-P	3	3.600 m	1.529 mm	0.00042	0.085	OK
KDS-ST-07-EX-P	4	3.600 m	0.942 mm	0.00026	0.052	OK
KDS-ST-07-EX-N	1	3.600 m	1.293 mm	0.00036	0.072	OK
KDS-ST-07-EX-N	2	3.600 m	1.856 mm	0.00052	0.103	OK
KDS-ST-07-EX-N	3	3.600 m	1.529 mm	0.00042	0.085	OK
KDS-ST-07-EX-N	4	3.600 m	0.942 mm	0.00026	0.052	OK
KDS-ST-07-EY-P	1	3.600 m	1.214 mm	0.00034	0.067	OK
KDS-ST-07-EY-P	2	3.600 m	1.695 mm	0.00047	0.094	OK
KDS-ST-07-EY-P	3	3.600 m	1.389 mm	0.00039	0.077	OK
KDS-ST-07-EY-P	4	3.600 m	0.851 mm	0.00024	0.047	OK
KDS-ST-07-EY-N	1	3.600 m	1.214 mm	0.00034	0.067	OK
KDS-ST-07-EY-N	2	3.600 m	1.695 mm	0.00047	0.094	OK
KDS-ST-07-EY-N	3	3.600 m	1.389 mm	0.00039	0.077	OK
KDS-ST-07-EY-N	4	3.600 m	0.851 mm	0.00024	0.047	OK
KDS-SVC-01	1	3.600 m	0.004 mm	0.00000	0.000	OK
KDS-SVC-01	2	3.600 m	0.004 mm	0.00000	0.000	OK
KDS-SVC-01	3	3.600 m	0.004 mm	0.00000	0.000	OK
KDS-SVC-01	4	3.600 m	0.015 mm	0.00000	0.001	OK
KDS-SVC-02	1	3.600 m	0.005 mm	0.00000	0.000	OK
KDS-SVC-02	2	3.600 m	0.006 mm	0.00000	0.000	OK
KDS-SVC-02	3	3.600 m	0.004 mm	0.00000	0.000	OK
KDS-SVC-02	4	3.600 m	0.019 mm	0.00001	0.001	OK
KDS-SVC-03-WX-P	1	3.600 m	0.456 mm	0.00013	0.025	OK
KDS-SVC-03-WX-P	2	3.600 m	0.582 mm	0.00016	0.032	OK
KDS-SVC-03-WX-P	3	3.600 m	0.426 mm	0.00012	0.024	OK
KDS-SVC-03-WX-P	4	3.600 m	0.251 mm	0.00007	0.014	OK
KDS-SVC-03-WX-N	1	3.600 m	0.456 mm	0.00013	0.025	OK
KDS-SVC-03-WX-N	2	3.600 m	0.582 mm	0.00016	0.032	OK

Combo	Story	Height	Drift	Ratio	D/L	Status
KDS-SVC-03-WX-N	3	3.600 m	0.426 mm	0.00012	0.024	OK
KDS-SVC-03-WX-N	4	3.600 m	0.251 mm	0.00007	0.014	OK
KDS-SVC-03-WY-P	1	3.600 m	0.515 mm	0.00014	0.029	OK
KDS-SVC-03-WY-P	2	3.600 m	0.635 mm	0.00018	0.035	OK
KDS-SVC-03-WY-P	3	3.600 m	0.461 mm	0.00013	0.026	OK
KDS-SVC-03-WY-P	4	3.600 m	0.265 mm	0.00007	0.015	OK
KDS-SVC-03-WY-N	1	3.600 m	0.515 mm	0.00014	0.029	OK
KDS-SVC-03-WY-N	2	3.600 m	0.635 mm	0.00018	0.035	OK
KDS-SVC-03-WY-N	3	3.600 m	0.461 mm	0.00013	0.026	OK
KDS-SVC-03-WY-N	4	3.600 m	0.265 mm	0.00007	0.015	OK
KDS-SVC-04-EX-P	1	3.600 m	1.294 mm	0.00036	0.072	OK
KDS-SVC-04-EX-P	2	3.600 m	1.857 mm	0.00052	0.103	OK
KDS-SVC-04-EX-P	3	3.600 m	1.529 mm	0.00042	0.085	OK
KDS-SVC-04-EX-P	4	3.600 m	0.945 mm	0.00026	0.053	OK
KDS-SVC-04-EX-N	1	3.600 m	1.294 mm	0.00036	0.072	OK
KDS-SVC-04-EX-N	2	3.600 m	1.857 mm	0.00052	0.103	OK
KDS-SVC-04-EX-N	3	3.600 m	1.529 mm	0.00042	0.085	OK
KDS-SVC-04-EX-N	4	3.600 m	0.945 mm	0.00026	0.053	OK
KDS-SVC-04-EY-P	1	3.600 m	1.215 mm	0.00034	0.067	OK
KDS-SVC-04-EY-P	2	3.600 m	1.695 mm	0.00047	0.094	OK
KDS-SVC-04-EY-P	3	3.600 m	1.389 mm	0.00039	0.077	OK
KDS-SVC-04-EY-P	4	3.600 m	0.853 mm	0.00024	0.047	OK
KDS-SVC-04-EY-N	1	3.600 m	1.215 mm	0.00034	0.067	OK
KDS-SVC-04-EY-N	2	3.600 m	1.695 mm	0.00047	0.094	OK
KDS-SVC-04-EY-N	3	3.600 m	1.389 mm	0.00039	0.077	OK
KDS-SVC-04-EY-N	4	3.600 m	0.853 mm	0.00024	0.047	OK

Result Postprocessing

Item	Value
Story rows	112
Member station rows	84
Foundation rows	9
Uplift nodes	0

Phase 3 Integrated Results

Item	Value
Version	p3-m19-integrated-results
Design items	197
Issue rows	34
Not checked count	0
Workflow locked	No
Ready for reviewer	No
Ticket coverage	Review
Nonlinear trace	p3-m16-nonlinear-trace
Gate	Review
Benchmark evidence	OK

Gate item	Value
Version	p3-m19-integrated-results-gate-v1
Tickets	P3-T58, P3-T59, P3-T61, P3-T62
Capacity points	13
Nonlinear steps	13
Method limitations	10
Approval state	not-submitted
Covered tickets	3/4

Ticket	Scope	Covered	Evidence
P3-T58	integrated-result-postprocessing	No	analysis=OK, 112 story rows, 84 member rows, 13 capacity points, 13 nonlinear steps, 197 design items, designReview=review-required, designIssues=34
P3-T59	calculation-report-method-limitations	Yes	10 method/limitation rows
P3-T61	workflow-lock	Yes	not-submitted / locked=false / editable=true / review=available
P3-T62	benchmark-regression	Yes	geometry:OK, hingeControl:OK, fiberNlth:OK

Module	Items	WARN	NG
rc	0	0	0
steel	84	0	0
connection	93	17	15

Module	Items	WARN	NG
foundation	20	1	0

Member Design Summary

Demand package	Value
Version	p2-design-demand-package
Active result	ENVELOPE
Member demands	84
Foundation demands	9

Member	Role	Status	Util.	Governing	Combo
M1	column	OK	0.292	steel-interaction	KDS-ST-05-EY-N
M2	column	OK	0.338	steel-interaction	KDS-ST-05-EX-N
M3	column	OK	0.292	steel-interaction	KDS-ST-05-EY-N
M4	column	OK	0.333	steel-interaction	KDS-ST-05-EY-N
M5	column	OK	0.381	steel-interaction	KDS-ST-05-EX-N
M6	column	OK	0.333	steel-interaction	KDS-ST-05-EY-P
M7	column	OK	0.292	steel-interaction	KDS-ST-05-EY-P
M8	column	OK	0.338	steel-interaction	KDS-ST-05-EX-P
M9	column	OK	0.292	steel-interaction	KDS-ST-05-EY-P
M10	column	OK	0.232	steel-interaction	KDS-ST-05-EY-N
M11	column	OK	0.268	steel-interaction	KDS-ST-05-EX-P
M12	column	OK	0.232	steel-interaction	KDS-ST-05-EY-N
M13	column	OK	0.271	steel-interaction	KDS-ST-05-EY-P
M14	column	OK	0.307	steel-interaction	KDS-ST-05-EX-P
M15	column	OK	0.271	steel-interaction	KDS-ST-05-EY-P
M16	column	OK	0.232	steel-interaction	KDS-ST-05-EY-P
M17	column	OK	0.268	steel-interaction	KDS-ST-05-EX-P
M18	column	OK	0.232	steel-interaction	KDS-ST-05-EY-P
M19	column	OK	0.326	steel-deflection	KDS-ST-05-EY-N
M20	column	OK	0.327	steel-deflection	KDS-ST-05-EX-N
M21	column	OK	0.326	steel-deflection	KDS-ST-05-EY-N
M22	column	OK	0.328	steel-deflection	KDS-ST-05-EY-P
M23	column	OK	0.330	steel-deflection	KDS-ST-05-EX-P
M24	column	OK	0.328	steel-deflection	KDS-ST-05-EY-N
M25	column	OK	0.326	steel-deflection	KDS-ST-05-EY-P
M26	column	OK	0.327	steel-deflection	KDS-ST-05-EX-N

Member	Role	Status	Util.	Governing	Combo
M27	column	OK	0.326	steel-deflection	KDS-ST-05-EY-P
M28	column	OK	0.391	steel-deflection	KDS-ST-05-EX-N
M29	column	OK	0.392	steel-deflection	KDS-ST-05-EX-N
M30	column	OK	0.391	steel-deflection	KDS-ST-05-EX-P
M31	column	OK	0.392	steel-deflection	KDS-ST-05-EY-P
M32	column	OK	0.394	steel-deflection	KDS-ST-05-EX-N
M33	column	OK	0.392	steel-deflection	KDS-ST-05-EY-P
M34	column	OK	0.391	steel-deflection	KDS-ST-05-EX-N
M35	column	OK	0.392	steel-deflection	KDS-ST-05-EX-P
M36	column	OK	0.391	steel-deflection	KDS-ST-05-EX-P
M37	beam	OK	0.166	steel-interaction	KDS-ST-05-EX-P
M38	beam	OK	0.142	steel-interaction	KDS-ST-05-EY-N
M39	beam	OK	0.166	steel-interaction	KDS-ST-05-EX-N
M40	beam	OK	0.142	steel-interaction	KDS-ST-05-EY-N
M41	beam	OK	0.142	steel-interaction	KDS-ST-05-EY-N
M42	beam	OK	0.166	steel-interaction	KDS-ST-05-EX-P
M43	beam	OK	0.142	steel-interaction	KDS-ST-05-EY-P
M44	beam	OK	0.166	steel-interaction	KDS-ST-05-EX-N
M45	beam	OK	0.142	steel-interaction	KDS-ST-05-EY-P
M46	beam	OK	0.142	steel-interaction	KDS-ST-05-EY-P
M47	beam	OK	0.166	steel-interaction	KDS-ST-05-EX-P
M48	beam	OK	0.166	steel-interaction	KDS-ST-05-EX-N
M49	beam	OK	0.161	steel-interaction	KDS-ST-05-EX-N
M50	beam	OK	0.161	steel-deflection	KDS-ST-05-EY-N
M51	beam	OK	0.161	steel-interaction	KDS-ST-05-EX-P
M52	beam	OK	0.163	steel-deflection	KDS-ST-05-EY-N
M53	beam	OK	0.161	steel-deflection	KDS-ST-05-EY-N
M54	beam	OK	0.161	steel-interaction	KDS-ST-05-EX-N
M55	beam	OK	0.161	steel-deflection	KDS-ST-05-EY-P
M56	beam	OK	0.161	steel-interaction	KDS-ST-05-EX-P
M57	beam	OK	0.163	steel-deflection	KDS-ST-05-EY-P
M58	beam	OK	0.161	steel-deflection	KDS-ST-05-EY-P
M59	beam	OK	0.161	steel-interaction	KDS-ST-05-EX-N
M60	beam	OK	0.161	steel-interaction	KDS-ST-05-EX-P

Member	Role	Status	Util.	Governing	Combo
M61	beam	OK	0.199	steel-deflection	KDS-ST-05-EX-N
M62	beam	OK	0.237	steel-deflection	KDS-ST-05-EY-N
M63	beam	OK	0.199	steel-deflection	KDS-ST-05-EX-P
M64	beam	OK	0.239	steel-deflection	KDS-ST-05-EY-N
M65	beam	OK	0.237	steel-deflection	KDS-ST-05-EY-N
M66	beam	OK	0.201	steel-deflection	KDS-ST-05-EX-N
M67	beam	OK	0.237	steel-deflection	KDS-ST-05-EY-P
M68	beam	OK	0.201	steel-deflection	KDS-ST-05-EX-P
M69	beam	OK	0.239	steel-deflection	KDS-ST-05-EY-P
M70	beam	OK	0.237	steel-deflection	KDS-ST-05-EY-P
M71	beam	OK	0.199	steel-deflection	KDS-ST-05-EX-N
M72	beam	OK	0.199	steel-deflection	KDS-ST-05-EX-P
M73	beam	OK	0.238	steel-deflection	KDS-ST-05-EX-P
M74	beam	OK	0.283	steel-deflection	KDS-ST-05-EY-P
M75	beam	OK	0.238	steel-deflection	KDS-ST-05-EX-N
M76	beam	OK	0.285	steel-deflection	KDS-ST-05-EY-P
M77	beam	OK	0.283	steel-deflection	KDS-ST-05-EY-P
M78	beam	OK	0.240	steel-deflection	KDS-ST-05-EX-P
M79	beam	OK	0.283	steel-deflection	KDS-ST-05-EY-N
M80	beam	OK	0.240	steel-deflection	KDS-ST-05-EX-N
M81	beam	OK	0.285	steel-deflection	KDS-ST-05-EY-N
M82	beam	OK	0.283	steel-deflection	KDS-ST-05-EY-N
M83	beam	OK	0.238	steel-deflection	KDS-ST-05-EX-P
M84	beam	OK	0.238	steel-deflection	KDS-ST-05-EX-N

Member Design Trace Matrix

Member	Type	Status	Formula rows	Action items
M1	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M2	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M3	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M4	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.

Member	Type	Status	Formula rows	Action items
M5	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M6	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M7	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M8	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M9	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M10	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M11	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M12	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M13	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M14	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M15	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M16	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M17	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M18	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M19	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M20	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M21	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M22	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M23	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M24	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.

Member	Type	Status	Formula rows	Action items
M25	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M26	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M27	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M28	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M29	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M30	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M31	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M32	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M33	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M34	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M35	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M36	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M37	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M38	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M39	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M40	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M41	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M42	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M43	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M44	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.

Member	Type	Status	Formula rows	Action items
M45	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M46	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M47	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M48	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M49	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M50	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M51	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M52	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M53	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M54	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M55	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M56	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M57	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M58	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M59	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M60	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M61	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M62	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M63	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M64	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.

Member	Type	Status	Formula rows	Action items
M65	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M66	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M67	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M68	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M69	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
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M71	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
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M73	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M74	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M75	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
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M79	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M80	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M81	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M82	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M83	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M84	steel	OK	9	Confirm compactness, lateral bracing, and connection design in project-specific checks.

Detailing And Foundation Summary

Scope	Rows	Warnings/NG	Max ratio
RC reinforcement	0	0	0.000
Steel review	84	0	0.394
Connection/foundation	93	23	2.567

Audit	Status
Load derivation attached	OK
Combination results available	OK
Member checks available	OK
RC schedule available	Not applicable
Steel schedule available	OK
Foundation review available	OK
Phase 3 integrated results available	OK
Phase 5 analysis cases listed	OK
Analysis run provenance records	Not applicable
Verified-only design transfer guard	OK
Phase 3 default not-checked cleanup	OK

Appendix

Practice Platform Readiness

Item	Value
Status	OK
Revision	R0
Approval	not-submitted
QA OK	Yes

Practice Validation

Item	Value
Status	WARN
P-Delta	WARN
Result tables	OK
Calculation trace	OK
Open issues	0

Remaining Design Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.
- Full KDS equation trace for every generated load combination and project-specific exception.
- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Independent reference-model correlation is not attached.
- Project-specific KDS inputs and final member selections require confirmation.
- This report is not a sealed structural calculation package.

Package Limitations

- Print-ready HTML is intended for browser PDF output.
- Final sealed calculation packages require project-specific engineering review.
- Unsupported checks remain listed in the appendix rather than hidden.